

JMO – 2019

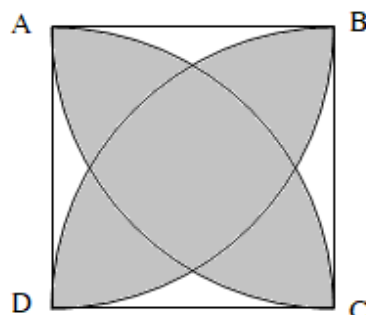
Answer all questions. All questions carry equal marks.

Full Marks 100

1. If the counting numbers x, y, z represents three sides of any triangle and $y < z$, then find the maximum number of triangles possible.
2. Find the sum of different 10^{th} place digits of all perfect square numbers whose unit's place digits are perfect square.
3. Arrange 16 plants in 8 rows with 4 plants in each row.
4. Find the remainder when 2^{2019} is divided by 2019.
5. If $x + y + z = 6$, $x^2 + y^2 + z^2 = 26$ and $x^3 + y^3 + z^3 = 90$ then find $x^4 + y^4 + z^4$.
6. Find the sum: $\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \dots + \frac{50}{1+50^2+50^4}$.
7. If $E = \left(\frac{1}{81}\right)^{-16^{-4^{-2^{-1}}}} + (25)^{-8^{-3^{-1}}}$
8. Find the positive 4 digit number whose value is increased by exactly 60% when the 1st digit from the left is shifted to the end of the number.
9. Between 5'O clock and 6'O clock a man looked at his watch and by mistake he read the hour hand as minute hand and vice versa. As a result he observed that the time was 57 minutes earlier than the correct time. Determine the correct time.
10. In a 100 m race, Ramesh defeated Shyam by 15 m. In another competition of a similar race Harish defeated Shyam by 20 m. If Ramesh and Harish taken part in 150 m race, then who will win the race by what distance.
11. A and B have some marbles with them. When A gives 100 marbles to B then the number of marbles with B becomes double that of A. However when B gives some marbles to A, then A will have 5 times more marbles than B. What was the minimum possible number of marbles with A originally and how many marbles were there with B in such case?
12. Five mens A, B, C, D and E are wearing caps of black or white colour without knowing the colour of their own caps. It is known that a man wearing a black cap always speaks the truth while the one wearing white always tells lies. If they make the following statement. Find the colour of cap worn by each of them.
A: I see three black caps and one white cap.
B: I see four white caps.
C: I see one black cap and three white caps
D: I see four black caps.
E: no statement.
13. Two women went to a market to sell 100 eggs. Out of 100 eggs one of them took some eggs and went to a different place to sell. After selling all the eggs each received the same amount. Then the first woman told: had I sold your eggs I would have got $66\frac{2}{3}$ rupees. Determine how eggs each woman sold?
14. ABCD is a quadrilateral in which $\angle DAB + \angle ABC = 90^\circ$. If $BC = 9$ cm, $BD = 6$ cm and $AC = 7$ cm, then find the area of square CDEF.
15. ABCDEF is a regular hexagon. Diagonals FC and BD intersect each other at G. Determine the ratio of the area of the triangle BCG and the area of the quadrilateral DEFG.

JMO QUESTIONS (ORISSA)

16. In the square ABCD, P is one point inside the square such that $PA = 15$ cm, $PC = 20$ cm, $PD = 7$ cm. Determine PB.
17. Area of triangle ABC is 14 cm^2 . D is a point on BC such that $CD = 3BD$, E is the midpoint of AD, CE when produced meets AB at F. Find the sum of the area of the triangles CDE and AEF.
18. In the figure ABCD is a square of side 10 cm each. Quadrants are drawn with each vertex of the square as centre. Find the area of the shaded portion. (Take $\sqrt{3} = 1.732$, $\pi = 3.14$)



19. In the $\triangle ABC$, $AB = 60$ cm, $AC = 80$ cm, $BC = 100$ cm and D is a point on BC such that the perimeter of $\triangle ABD$ and perimeter of $\triangle ACD$ are equal. Find AD.
20. Solve: $\frac{1}{x} + \frac{1}{y+z} = \frac{1}{2}$, $\frac{1}{y} + \frac{1}{z+x} = \frac{1}{3}$, $\frac{1}{z} + \frac{1}{x+y} = \frac{1}{4}$

